SP250 Homework 7

1. 6 students arrive at the university.

   a) (3 points): How many ways are there for them to break into two teams of three and play some beach volleyball while they wait for their rooms to be ready?
   b) (3 points): Rooms are now ready. How many ways are there to assign students to rooms if each room should have exactly two students? (for the avoidance of doubt, the rooms are distinguishable).
   c) (3 points): How many ways are there to break these 6 students into 3 study groups of 2 students each?
   d) (1 point, harder): How many ways are there to break these 6 students into 3 study groups so that each study group is not empty?

Solution:

---

2. Yijing wrote 11 numbers on the whiteboard: six zeros and five ones. He then performs the following operation 10 times: erase any two numbers. If they were equal, write a 0 on the whiteboard. If they were not equal, write a 1. Prove that the last remaining number does not depend on the order in which Yijing does his operations and find what it is.

Solution:

---

3. In a country of Vowelia, all words satisfy all of the following conditions:
   - Have at most 3 letters.
   - Each letter is one of the letters in the word Vowelia.
   - Each letter is used at most once.
   - Each word contains at least one consonant and at least one vowel. For the avoidance of doubt, consonants and vowels in Vowelia are the same as in English language.

   How many distinct words are there in Vowelia?

Solution:

---

4. How many 4-digit positive integers have four different digits, where the leading digit is not zero, the integer is a multiple of 5, and 5 is the largest digit? (Friendly reminder: check your answer).

Solution:

5. A) A grasshopper jumps along a line. Every second, it jumps 1,3, or 5 yards either to the left or to the right. Is it possible that the grasshopper ended up at the starting point after exactly 2015 hops?

B) A different grasshopper jumps along a line. Every second it jumps either 1 yard to the left or 2 yards to the right. (Only these two jumps are possible). Is it possible that the grasshopper ended up at the starting point after exactly 2015 hops?

Solution:

---

6. (harder): Prove that among any 10 integers, one can pick several, whose sum is a multiple of 10. (For the avoidance of doubt, several means more than zero).

Solution:
